GLOBAL POLLUTION ANALYSIS AND ENERGY RECOVERY

Machine Learning Based Classification and Environmental Analysis

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Dataset	Global Pollution Analysis
Models	Multinomial Naive Bayes, KNN, Decision Tree

This report presents the preprocessing, classification, evaluation and energy-recovery analysis carried out on a 200-record global pollution dataset.

1. Introduction

Pollution is influenced by several environmental and human activities, including industrial waste, energy consumption, emissions and waste generation. Looking at these factors together can help in understanding pollution conditions and identifying areas where environmental management can be improved.

In this project, country-year observations are classified into three pollution severity categories: Low, Medium and High. Multinomial Naive Bayes, K-Nearest Neighbors (KNN), and Decision Tree models are used for classification. The project also studies energy recovery in relation to industrial waste, renewable energy and pollution-related variables.

2. Objectives

- Clean and prepare the global pollution dataset for analysis.
- Calculate a Pollution Index and divide observations into Low, Medium and High severity categories.
- Prepare environmental, energy, economic and demographic features for classification.
- Implement Multinomial Naive Bayes, KNN and Decision Tree classifiers.
- Tune KNN and Decision Tree parameters using 5-fold cross-validation.
- Evaluate the models using accuracy, precision, recall, F1-score and confusion matrices.
- Study the relationship between pollution and energy recovery.
- Present useful findings and practical recommendations.

3. Dataset Description

The dataset contains 200 observations and 13 original columns. It includes country and year information along with air, water and soil pollution indices, industrial waste, recovered energy, CO■ emissions, renewable energy share, plastic waste, energy consumption per capita, population and GDP per capita.

Aspect	Result
Observations	200
Original features	13
Missing values	0
Duplicate rows	0
Severity classes	Low, Medium, High

4. Data Preprocessing

The dataset was checked for missing values, duplicate records and data types. The original dataset had no missing values and no duplicate rows. Median imputation was included in the workflow so that the notebook can also handle missing numerical values if they occur.

An overall Pollution Index was calculated as the average of the Air Pollution Index, Water Pollution Index and Soil Pollution Index. Quantile-based grouping produced 67 Low, 66 Medium and 67 High observations.

Country and year were encoded during preprocessing. Country encoding was not used as a numerical predictor because the assigned numbers would create an artificial ordering. The yearly pollution trend was analyzed separately.

A major step was avoiding target leakage. Since the Pollution Index and the three individual pollution indices define the target, Air_Pollution_Index, Water_Pollution_Index and Soil_Pollution_Index were not used as model inputs. The classifiers instead used other environmental, energy, economic and demographic variables.

Features Used for Classification

- Year
- Industrial Waste (in tons)
- Energy Recovered (in GWh)
- CO■ Emissions (in MT)
- Renewable Energy (%)
- Plastic Waste Produced (in tons)
- Energy Consumption Per Capita (in MWh)
- Population (in millions)
- GDP Per Capita (in USD)

5. Classification Methodology

The data was divided into training and testing sets using an 80:20 stratified split. The training set contained 158 observations and the test set contained 42 observations, with 14 observations from each severity class in the test set.

Outlier limits were calculated from training data only using the IQR method. The same limits were applied to the test data. Feature scaling was also fitted using the training data before being applied to both sets.

5.1 Multinomial Naive Bayes

A manual Multinomial Naive Bayes implementation was used. It estimates class probabilities from the scaled feature values and assigns each observation to the class with the highest resulting score.

5.2 K-Nearest Neighbors

KNN classifies an observation from its nearest training observations. K values from 1 to 15 were tested and the final value was selected using 5-fold cross-validation on the training data only. The selected K was 7, with a cross-validation accuracy of 0.392.

5.3 Decision Tree

The Decision Tree uses Gini impurity to split the data. max_depth and min_samples_split were tuned using 5-fold cross-validation. The selected parameters were max_depth = 7 and min_samples_split = 5, with a cross-validation accuracy of 0.380.

6. Model Evaluation Results

A majority-class baseline was included as a simple reference. Because the three target classes are almost equally balanced, the baseline accuracy was approximately one-third.

Model	Accuracy	Precision	Recall	F1 Score
Baseline	0.333	—	—	—
Multinomial Naive Bayes	0.238	0.221	0.238	0.227
KNN	0.357	0.372	0.357	0.358

Decision Tree	0.262	0.264	0.262	0.262

KNN produced the highest test accuracy among the three classifiers at 0.357. Its weighted precision and F1-score were 0.372 and 0.358. However, the accuracy was only slightly above the 0.333 baseline. This suggests that the non-leakage features retained for the models have limited ability to separate the three severity classes in this dataset.

Naive Bayes and Decision Tree performed below the baseline on the final test set. The result should be interpreted as a limitation of the available predictors and dataset rather than simply as a model failure.

7. Energy Recovery Analysis

The project also examined the connection between pollution-related conditions and energy recovery. Industrial waste was compared with recovered energy, while CO■ emissions and renewable energy share were examined against recovered energy.

Analysis	Observed result
Highest average pollution index in Top-10 analysis	Saint Barthelemy – 187.04
Severity with highest average recovered energy	Medium – 277.09 GWh
Severity with highest average renewable energy share	Low – 28.15%

These observations are descriptive and should not be treated as proof of cause and effect. They mainly show patterns in the available data that can be explored further with larger datasets.

8. Key Findings

- The three severity classes are almost perfectly balanced: 67 Low, 66 Medium and 67 High observations.
- The three pollution indices used to construct the target were excluded from model predictors to avoid target leakage.
- KNN selected K = 7 using 5-fold cross-validation and produced the strongest test-set performance among the three models.
- KNN was only slightly above the balanced baseline, showing that the remaining predictors have limited classification strength.
- The Decision Tree selected max_depth = 7 and min_samples_split = 5 through cross-validation, but its test accuracy was below the baseline.
- Industrial waste, renewable energy, emissions and recovered energy are useful variables to consider together when discussing environmental management.

9. Recommendations

- Improve monitoring and management of industrial waste in areas showing higher pollution levels.
- Explore suitable waste-to-energy technologies where industrial waste can be safely converted into useful energy.
- Increase the use of renewable energy sources as part of cleaner energy strategies.
- Monitor air, water and soil pollution together instead of relying on a single indicator.
- Use more detailed environmental and economic data in future studies to improve pollution-severity classification.

- Use longer country-level time-series data in future work to study changes in pollution and energy recovery more effectively.

10. Limitations

The dataset contains only 200 observations, which limits the amount of information available for training and testing. Also, the target is constructed from the pollution indices that are deliberately excluded from the predictors to avoid leakage. The resulting classification task therefore asks whether the other environmental, energy, economic and demographic variables can explain the severity categories.

The results should be treated as an analysis of this dataset rather than a general prediction system for all countries. A larger dataset with more years and additional environmental indicators would provide a stronger basis for future modelling.

11. Conclusion

This project combined data preprocessing, machine learning classification and energy-recovery analysis to study global pollution conditions. The dataset was cleaned, a Pollution Index was calculated, and observations were divided into Low, Medium and High severity categories.

Three classification methods were implemented and evaluated. KNN gave the highest test accuracy of the three models, although its performance was only slightly above the balanced baseline. This indicates that, after removing direct target leakage, the available environmental, energy, economic and demographic variables do not strongly separate the three severity classes in this dataset.

The energy analysis also showed why industrial waste, renewable energy, emissions and recovered energy are useful to study together. Overall, the project provides a practical starting point for further environmental and energy-management analysis.